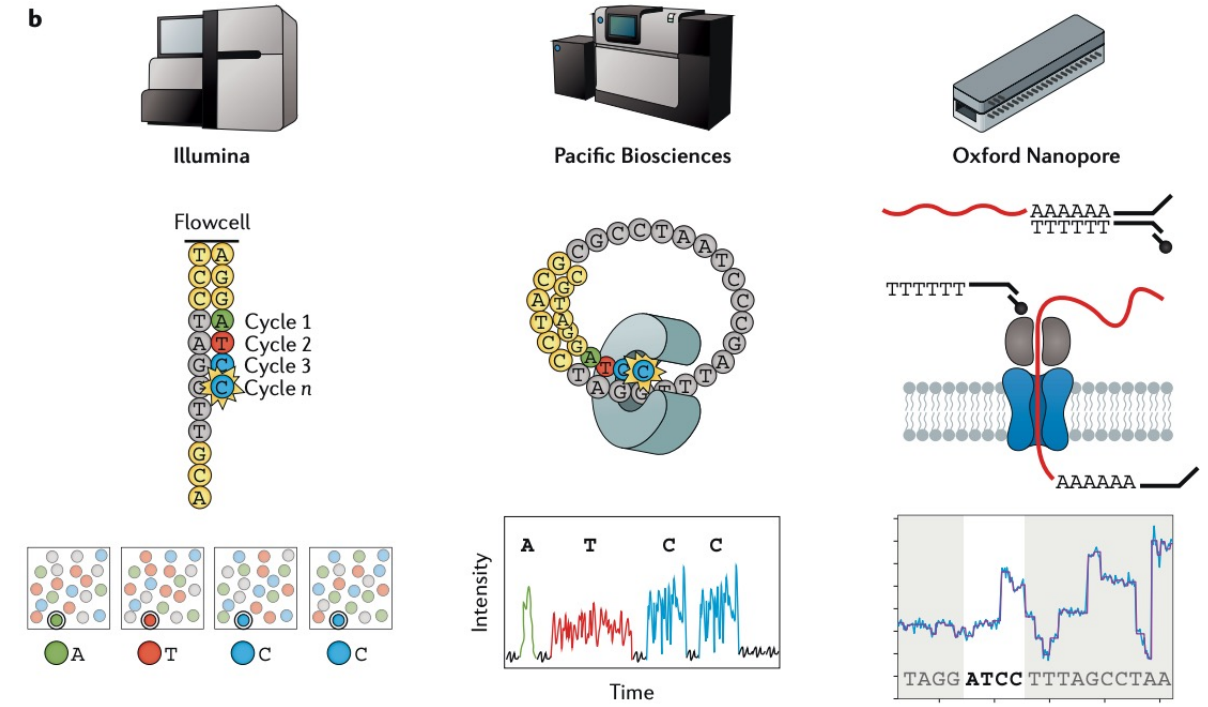


The RNA-seq Protocol

Erin Osborne Nishimura
DSCI 512:
RNA-seq data analysis
November 9, 2026

Genomics

Three flavors of high-throughput sequencing



Short read
NextGen

Long read
Third Gen

Long read or direct read
Third Gen

https://www.nature.com/articles/s41576-019-0150-2.pdf

Genomics

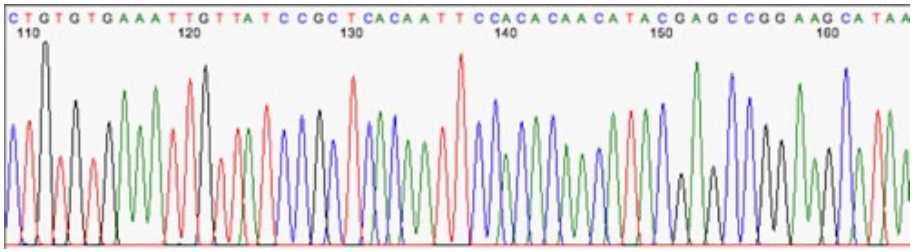
Three flavors of high-throughput sequencing

Short read
NextGen

Long read
Third Gen

Long read or direct read
Third Gen

- **Sanger.** “Old school”, first gen sequencing. 800 bp – 1 kb.

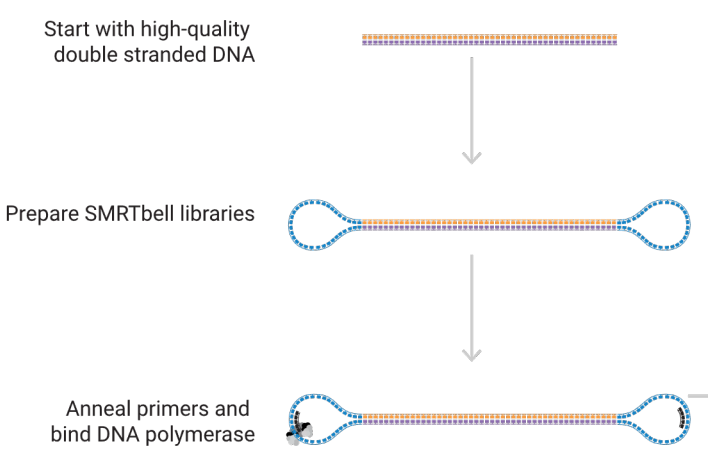


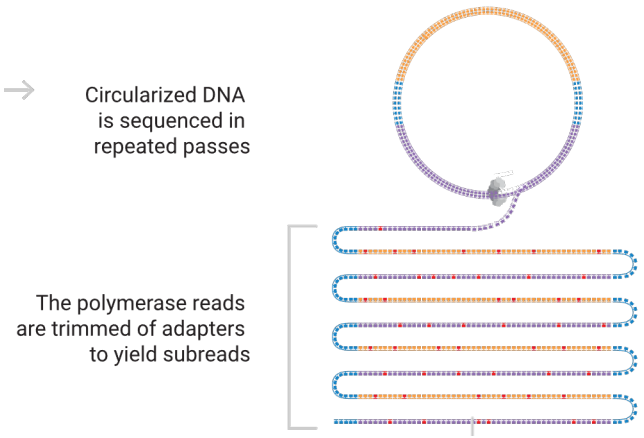
- **Short reads.** Illumina (aka **NextGen** sequencing). Whole genome.

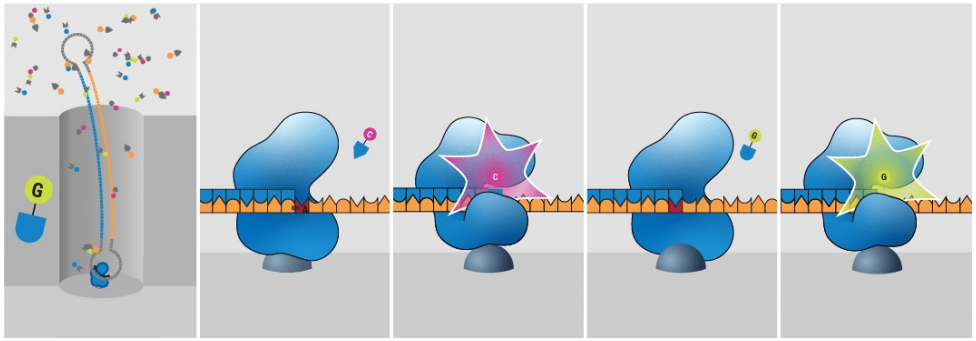
- **Third Gen sequencing.** Long reads. Pac Bio & minION (Oxford Nanopore Technologies). Whole genome.

Pac Bio

Circular long-read sequencing







- Immobilized DNA Polymerases at the bottom of each well amplify a single circularized DNA molecule.
- Colored light is emitted for each nucleotide
- Long reads are produced (~15,000 bp mean)
- Errors are minimized by going around each circle multiple times (90% reads > Q30)

Pac Bio Sequencing

A video to explore on your own



https://www.youtube.com/watch?v=7yYPHatccgw

MinION

Oxford Nanopore



Why MinION?

Active temperature control

Robust sequencing performance in a wide range of environments (10–35°C).

Sleek new design

Redesigned architecture with status LEDs for at-a-glance run status.

Palm sized

Sequence in the lab or field.

Real-time data

Immediate access to actionable insights.

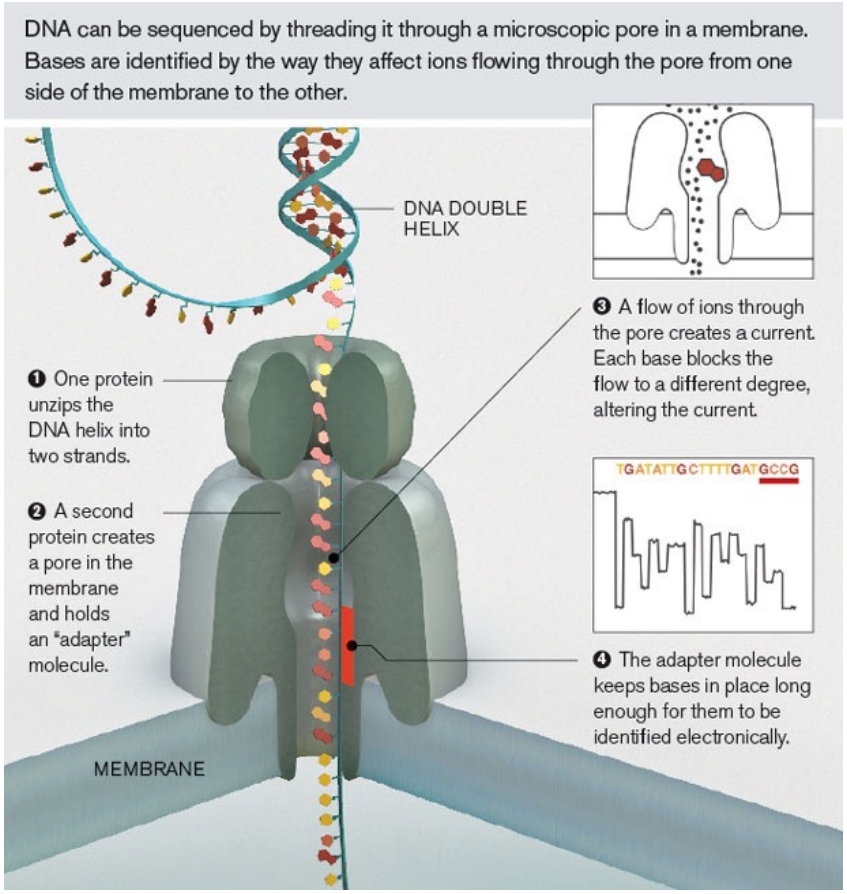
Any read length

Sequence native DNA or RNA fragments, from short to ultra-long, with built-in gold-standard methylation calling.

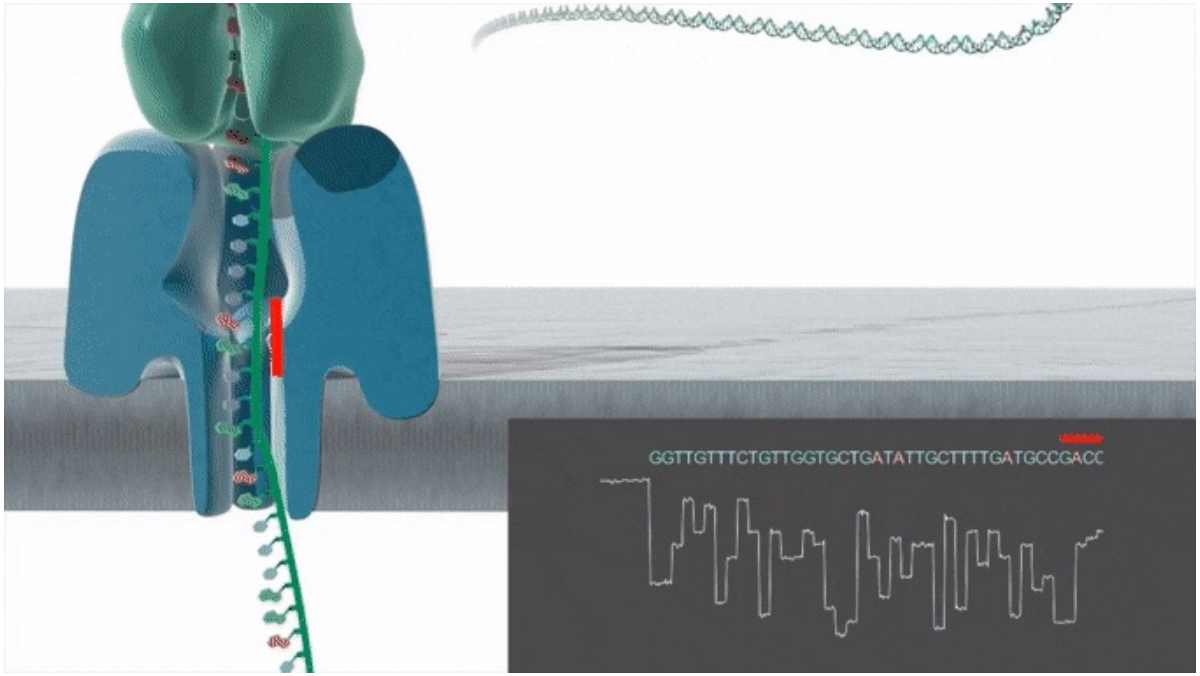
Oxford Nanopore

A third-generation sequencing technology

- minION
- A single, long, unamplified DNA molecule is dissociated into a single strand and fed through a small protein pore in a membrane.
- As the DNA passes through the membrane, a small current is created that is distinct for each nucleotide.
- The ionic flow signature of each nucleotide can be read through an extremely sensitive detector

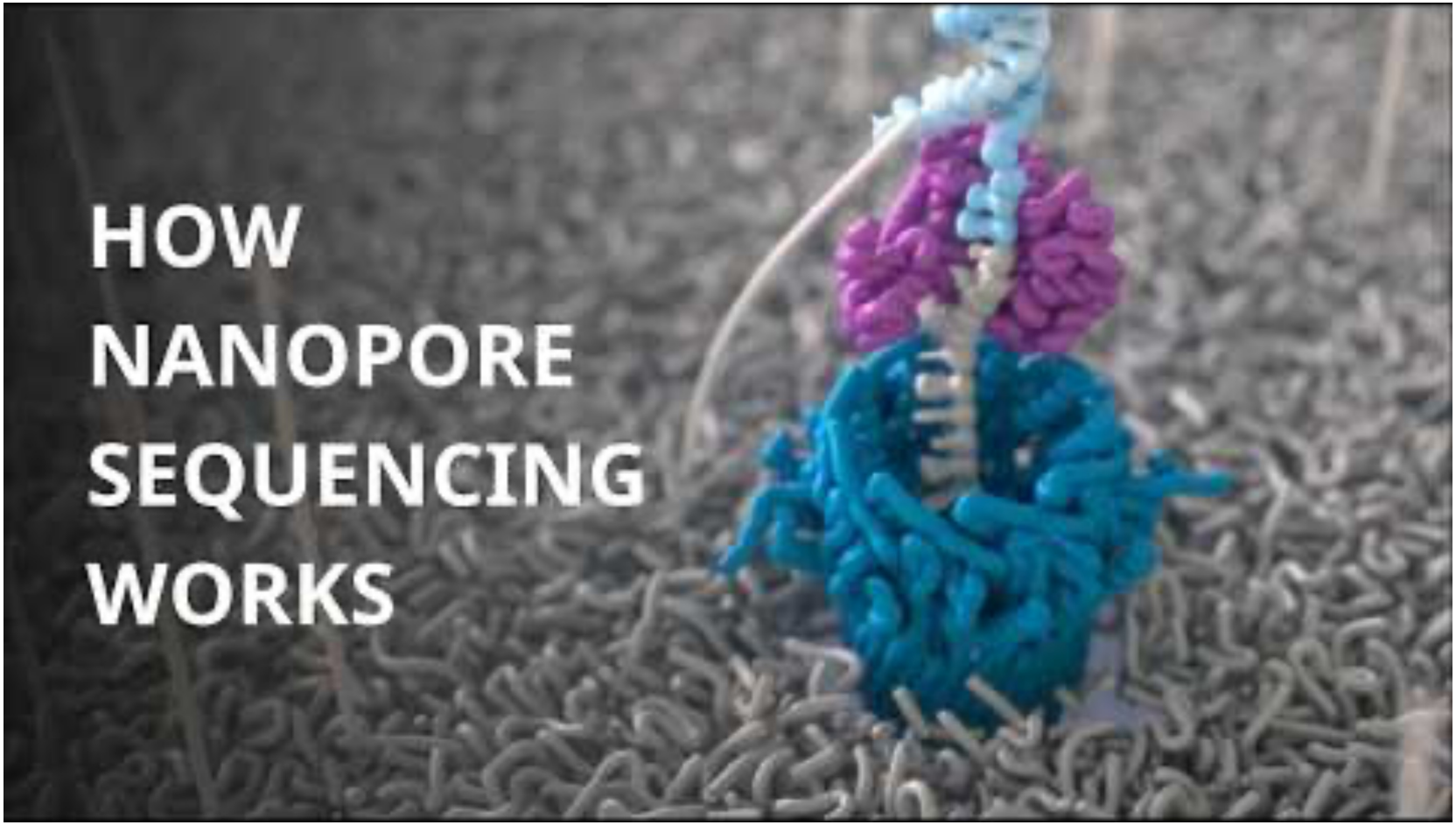


Nanopore sequencing technology



Nanopore Sequencing

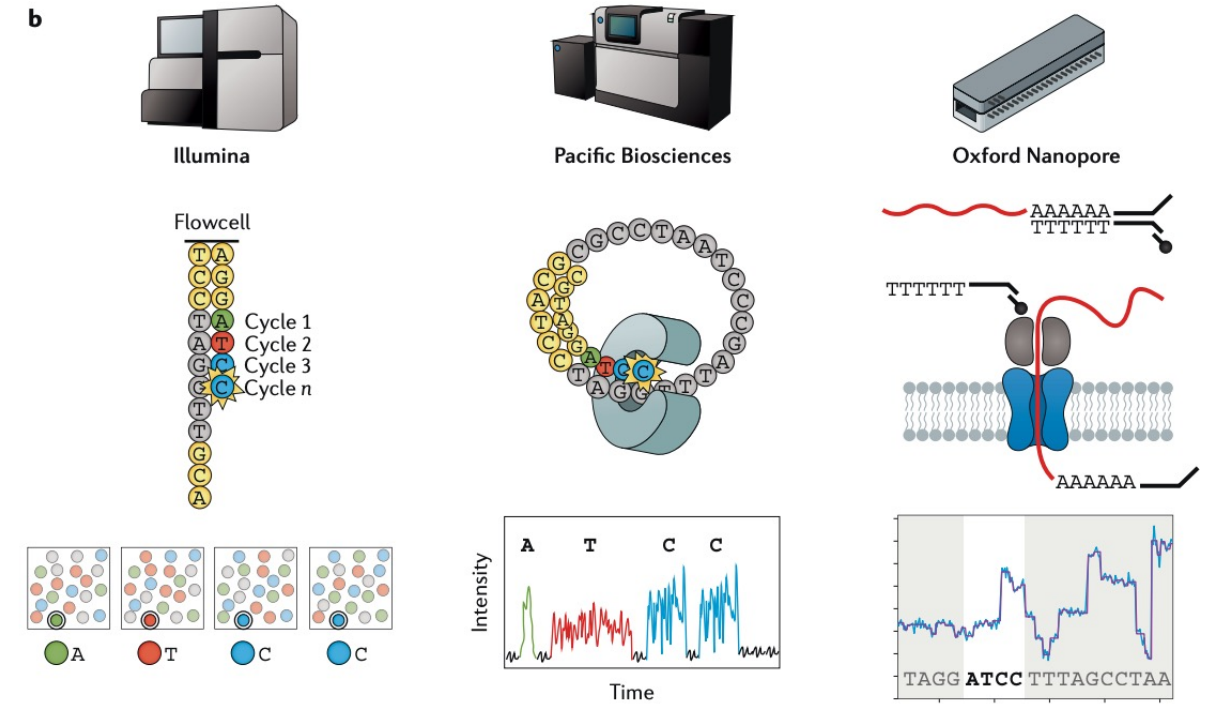
A video to explore on your own



https://youtu.be/VxGliKyYuFQ

Genomics

Three flavors of high-throughput sequencing



Short read
NextGen

Long read
Third Gen

Long read or direct read
Third Gen

https://www.nature.com/articles/s41576-019-0150-2.pdf

Illumina

Benchtop Next-Generation Sequencing



Key specifications	New  MiSeq i100 Series	 NextSeq 550 System	 NextSeq 1000 and 2000 Systems
Max output per flow cell	30 Gb [a]	120 Gb [b]	540 Gb
Run time (range)[c]	~4–24 hr	~11–29 hr	~8–44 hr
Max reads per run (single reads)	100M [a]	400M [b]	1.8B
Max read length	2 × 500 bp	2 × 150 bp	2 × 300 bp

Anschutz

Illumina

Production Next-Generation Sequencing



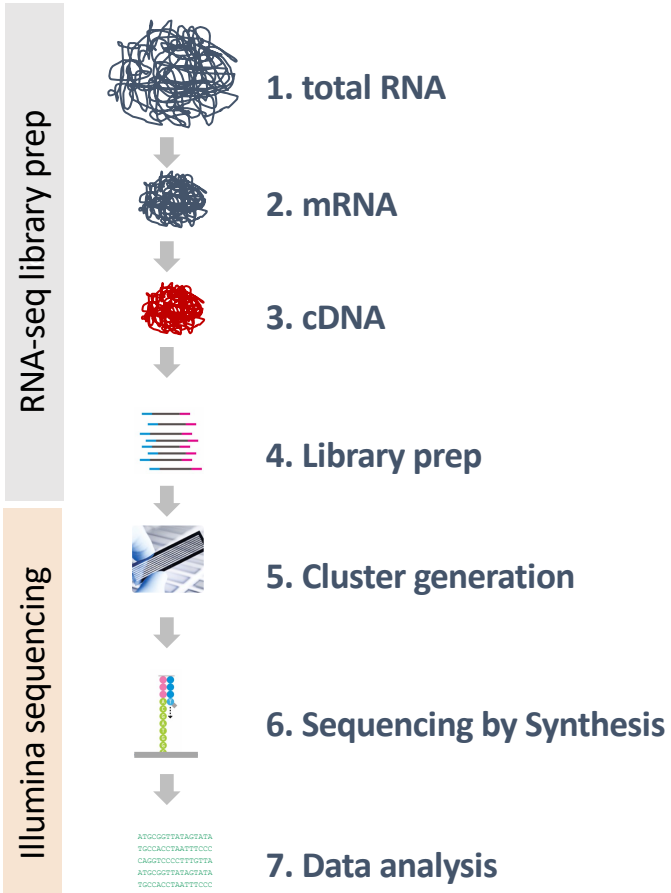
Key specifications	 NextSeq 1000 and 2000 Systems	 NovaSeq 6000 System	 NovaSeq X Series
Max output per flow cell	540 Gb[a]	3 Tb[b]	8–10.5 Tb[c]
Run time (range)[d]	~8–44 hr	~13–44 hr	~17–48 hr
Max reads per run (single reads)	1.8B[a]	10B (single flow cell)[b] 20B (dual flow cells)	26–35B (single flow cell)[c] 52–70B (dual flow cells)[c,e]
Max read length	2 × 300 bp	2 × 250 bp	2 × 300 bp

Anschutz

Anschutz

Illumina RNA-seq

Basic protocol overview



- Extract total RNA
- Enrich for mRNA
- Convert RNA to cDNA using Reverse Transcriptase
- Prepare libraries (fragmentation + adapter ligation)
- NEXTGEN sequencing is done at a facility. Samples are loaded onto a flow cell
- Sequencing by synthesis is performed to generate the sequence
- Data is sent back to researcher for alignment, quantification, and downstream analysis

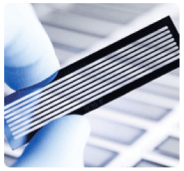
Illumina RNA-seq

A closer look



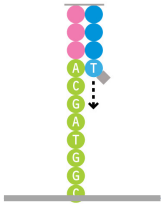
1-4. Library preparation

- create DNA fragments
- ligate adapters onto fragments



5. Cluster generation

- adhere fragments to a flow cell
- amplify them



6. Sequencing by synthesis

- directly image a specialized polymerization reaction that allows visualization of each nucleotide incorporated
- Tens of millions of fragments are sequenced

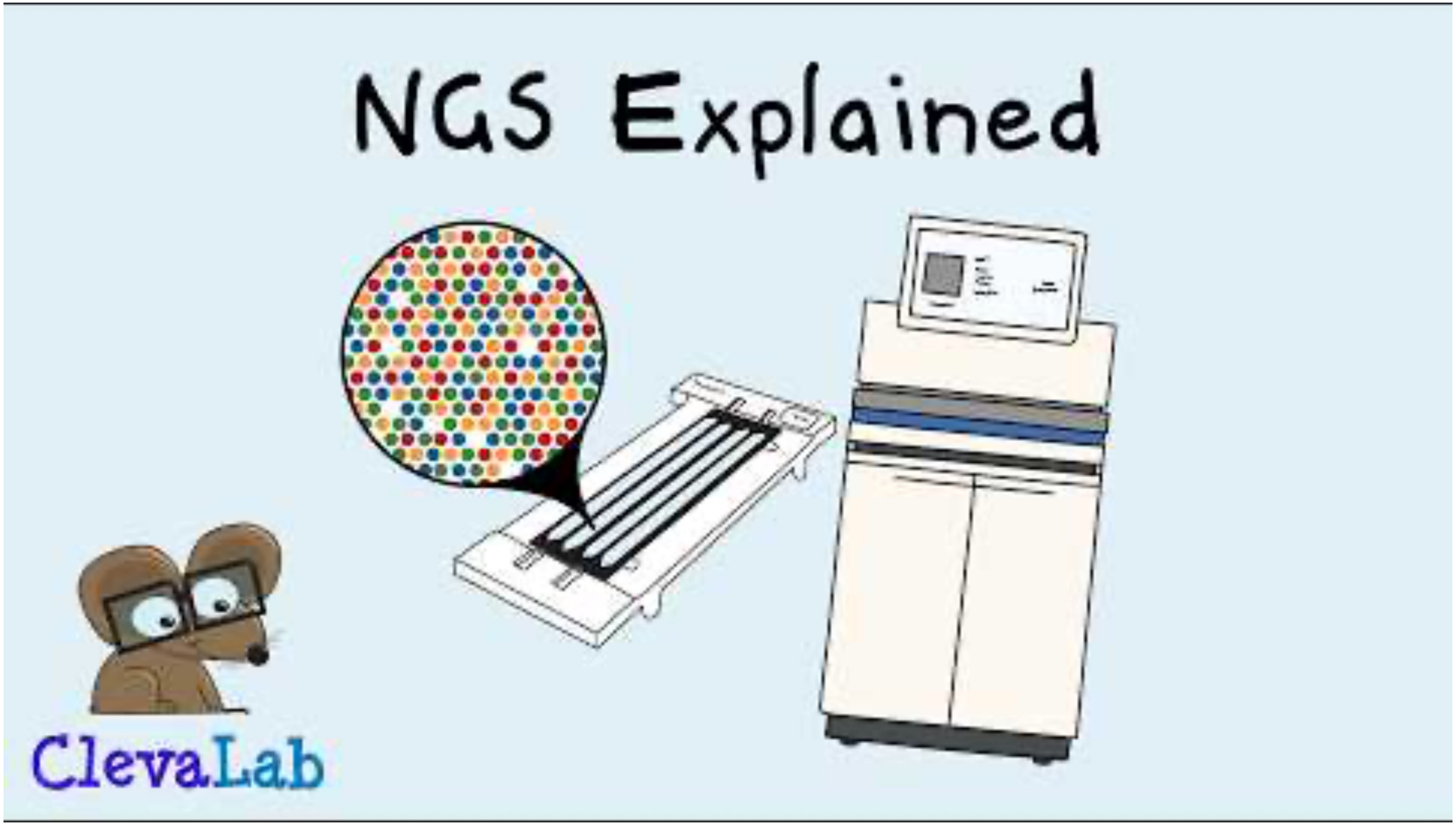
ATGCGGTTATAGTATA
TGCCACCTAATTTCCC
CAGGTCCCCTTTGTTA
ATGCGGTTATAGTATA
TGCCACCTAATTTCCC

7. Data analysis

- Use computer algorithms to align sequences to the genome

Illumina RNA-seq

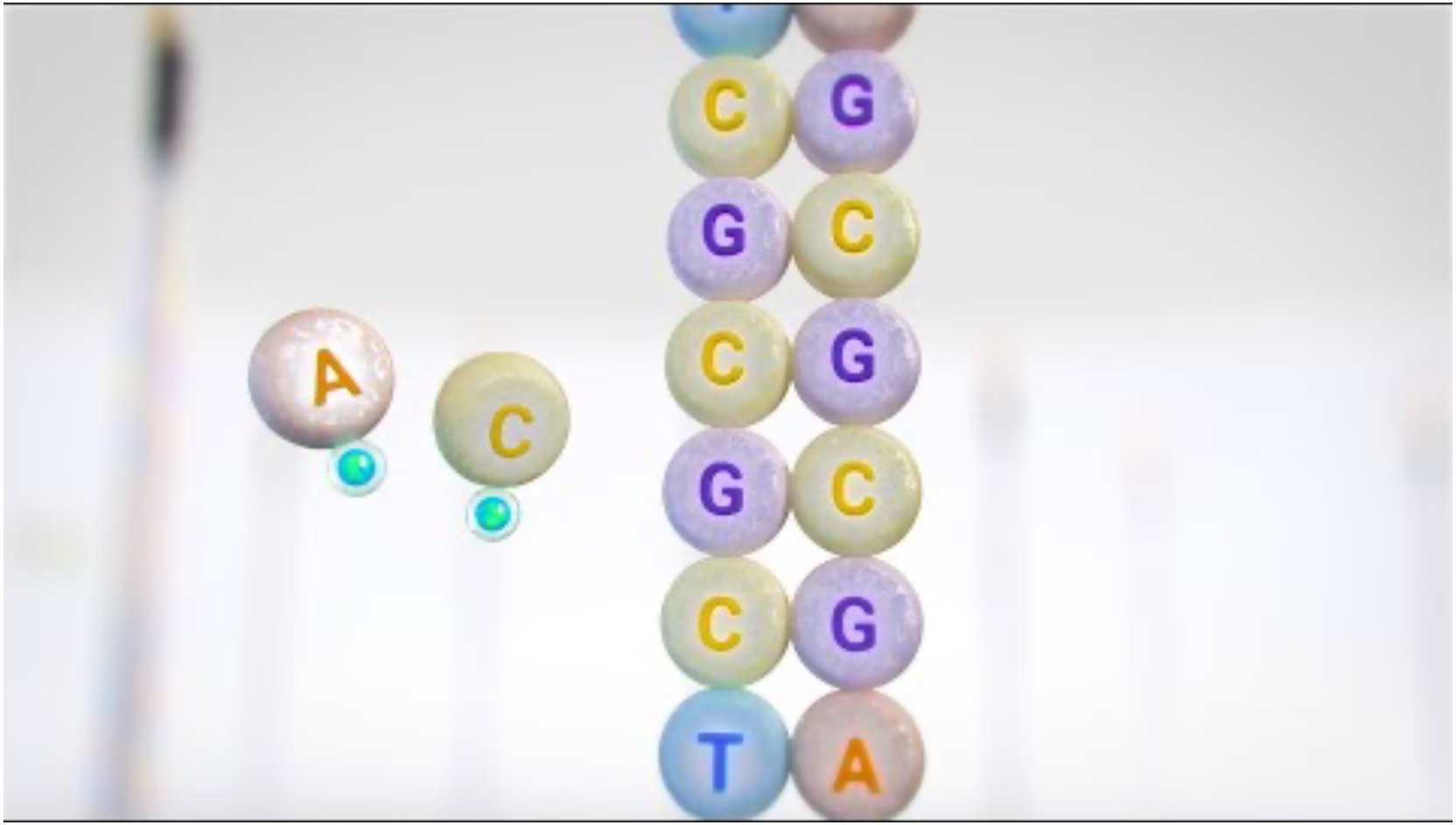
A video to watch in class



https://www.youtube.com/watch?v=WKAUtJQ69n8

Illumina RNA-seq

A video to explore on your own



https://www.youtube.com/watch?v=EDVKxSNdSic

Some key details

How do I know how much RNA I made?

Measure conc. w/a fluorometer

RNA-seq library prep

Illumina sequencing

1. total RNA
2. mRNA
3. cDNA
4. Library prep
5. Cluster generation
6. Sequencing by Synthesis
7. Data analysis

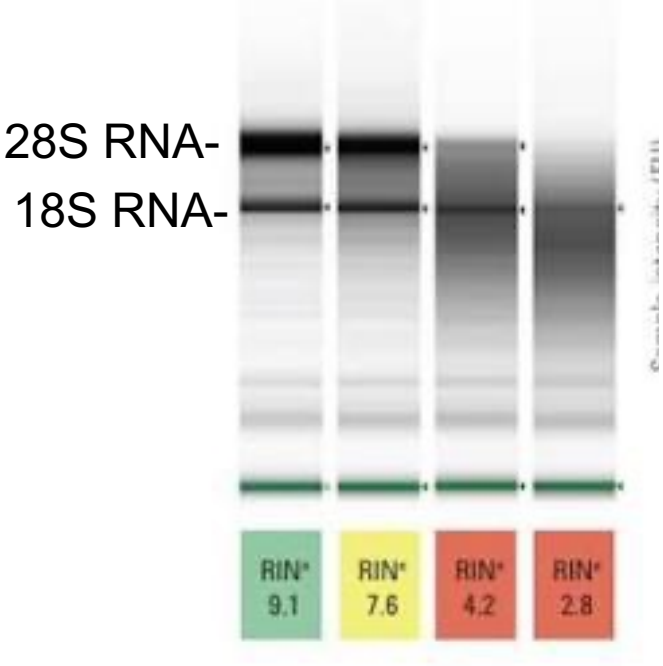


- Measure the concentration using a fluorometer
- Fluorometer v. Spectrometer
- Qubit or Promega quantus for example
- High specificity for RNA
- Sensitive
- Don't use a nanodrop!
- You don't need much RNA for RNA-seq!

Is my RNA good?

Look at size with a Tapestation





- Tapestation or Bioanalyzer – DNA, RNA, protein
- Analogous to running a gel but instead of running molecules through block of agar, they travel through a skinny capillary or “straw”
- Calculates RIN = RNA Integrity Number
- 10 high
- 1 low
- Should be > 8 for optimal RNA-seq
- Degraded RNA can still be sequenced! Use a special kit. Keep caveats in mind.

Where can I find these instruments?

Core Facilities Are Your Friends

Colorado State University Genomics Core

- Vet School on Drake Rd.
- Benchtop Sequencers, Tapestation, Qubit

Molecular Quantification Core

- Microbiology Building, Carol Wilusz
- Tapestation, Qubit

CU Anschutz Genomics Shared Resource

- Production Sequencers
- Tapestation, Qubit
- Single-cell RNA-seq

Erin's Lab & Other Labs

- Anatomy Zoology Building
- Tapestation, Qubit



Selecting mRNA

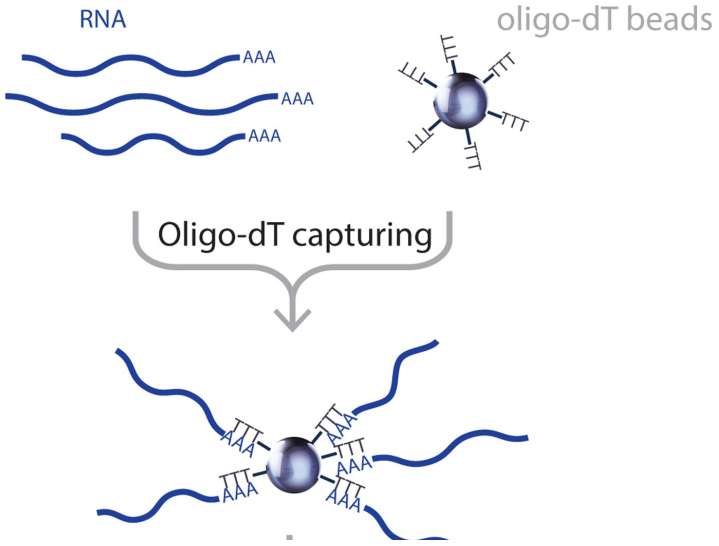
Lots of options

RNA-seq library prep

Illumina sequencing



- PolyA select with Oligo dT beads (cheapest & most popular)
- rRNA Deplete (RiboZero, RiboMinus, RiboGone)
- rRNA depletion is pricey, some kits are organism-specific, some rRNA is left, but it allows for non-polyA tailed RNA sequencing



How to library prep

Lots of kits

RNA-seq library prep

Illumina sequencing

1. total RNA
2. mRNA
3. cDNA
4. Library prep
5. Cluster generation
6. Sequencing by Synthesis
7. Data analysis

- NEB Next Kits (New England Biolabs)
- Watchmaker Genomics (CMB + BMB Alumni)
- $30 - $80 per sample
- Lots of options

WATCHMAKER GENOMICS



Adding adapters

A closer look

Dual index
(unique or combinatorial)



xGen UDI-UMI adapter



Flow cell binding sequence: Platform-specific sequences for library binding to instrument

Sequencing primer sites: Binding sites for general sequencing primers

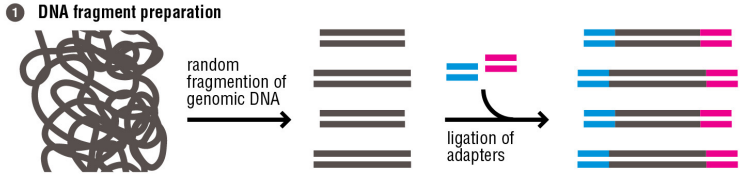
Sample indexes: Short sequences specific to a given sample library

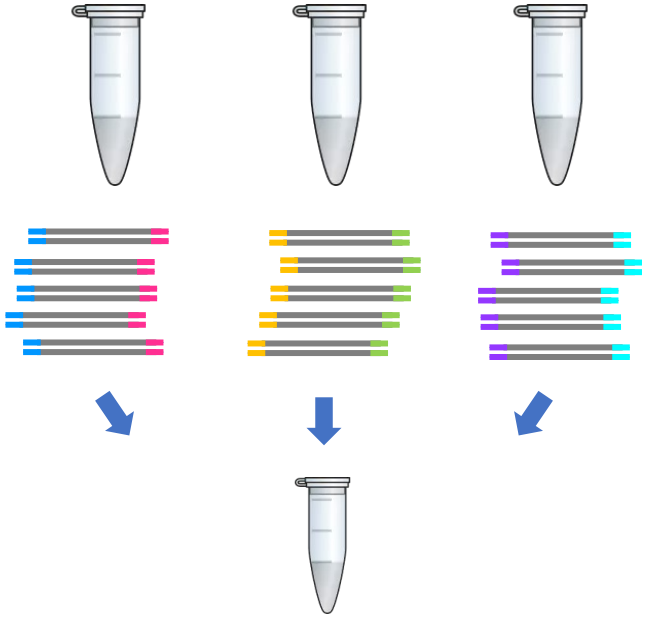
Molecular index/barcode: Short sequence used to uniquely tag each molecule in a given sample library

Insert: Target DNA or RNA fragment from a given sample library

Adding adapters

Multiplexing

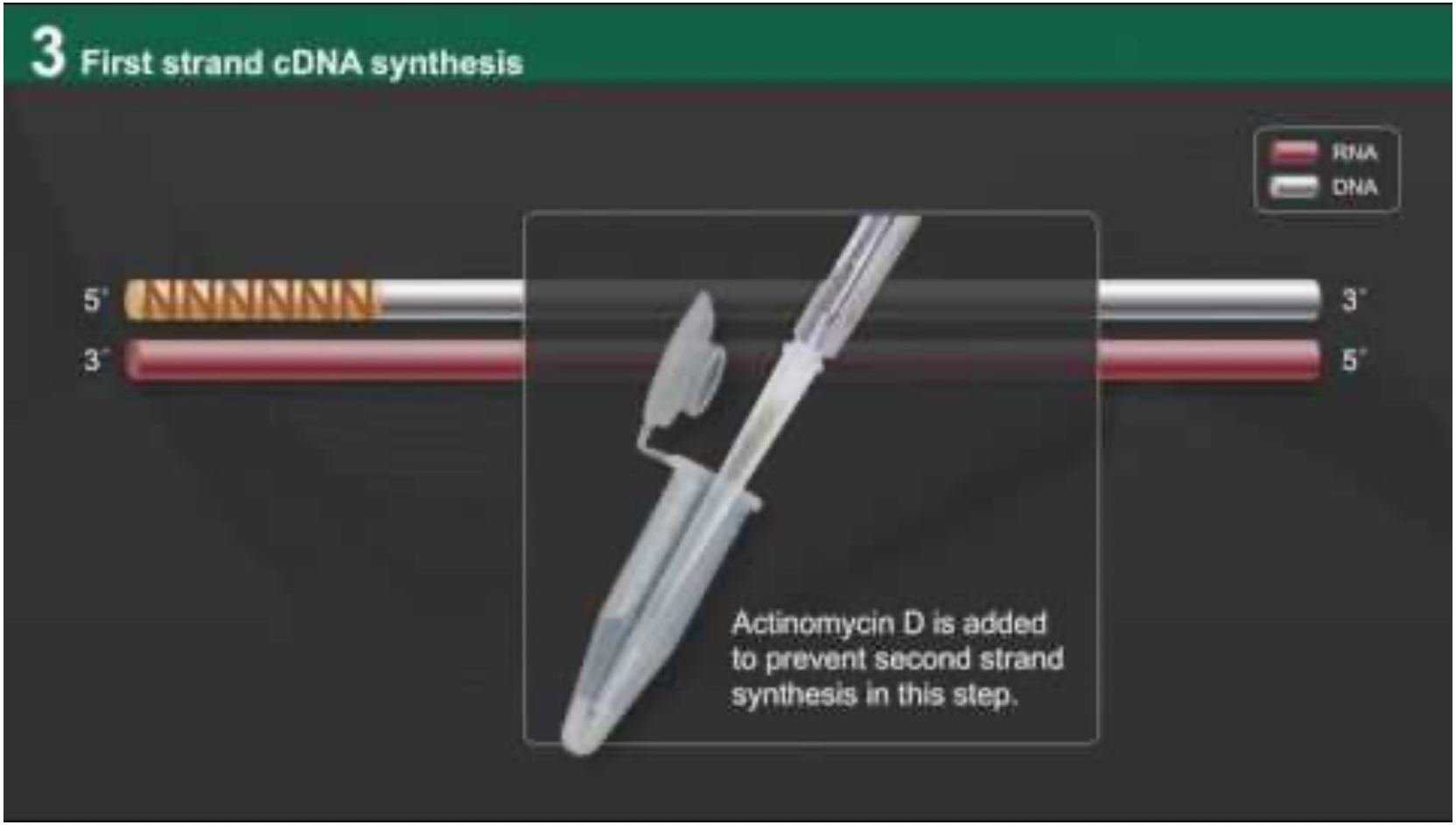




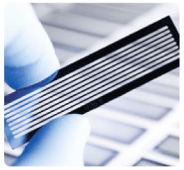
- Each sample is ligated to a specific adapter pair with its own sequence
- Samples are then are merged and run on the same flow cell (Up to 200 per flow cell)
- Samples are split computationally using unique adapter sequences

NEB Next Library Prep

A video to explore on your own



https://www.youtube.com/watch?v=OoBITqzcy1Y&t=3s

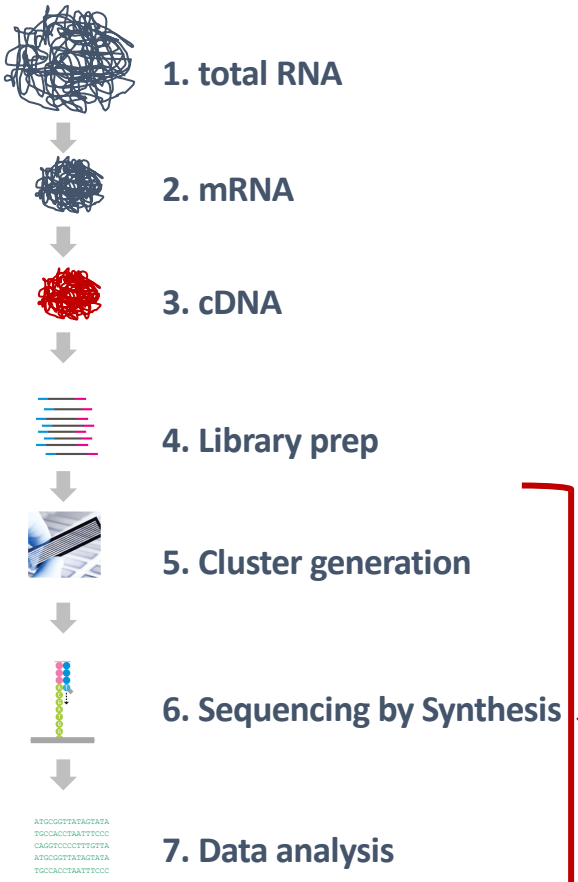


Who will sequence my samples?

Cores and Companies

RNA-seq library prep

Illumina sequencing

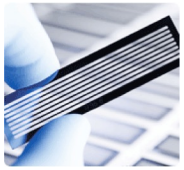


- Core Facilities (Anschutz, CSU Genomics Core)
- Currently, a wide array of companies
- Send them total RNA
- Support for some analysis

PLASMIDSAURUS

GENEWIZ® From Azenta Life Sciences

Novogene



What happens at the sequencing facility?

RNA-seq library prep

Illumina sequencing



1. total RNA



2. mRNA



3. cDNA



4. Library prep



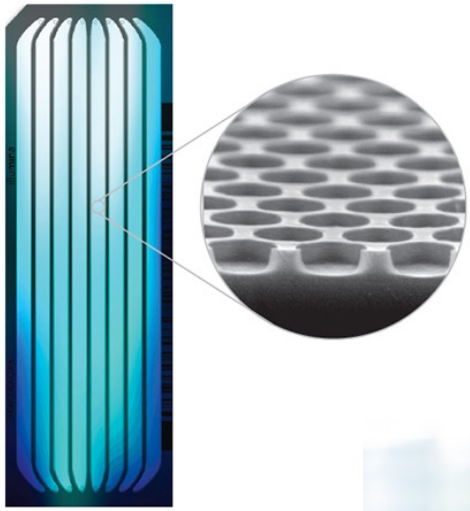
5. Cluster generation



6. Sequencing by Synthesis



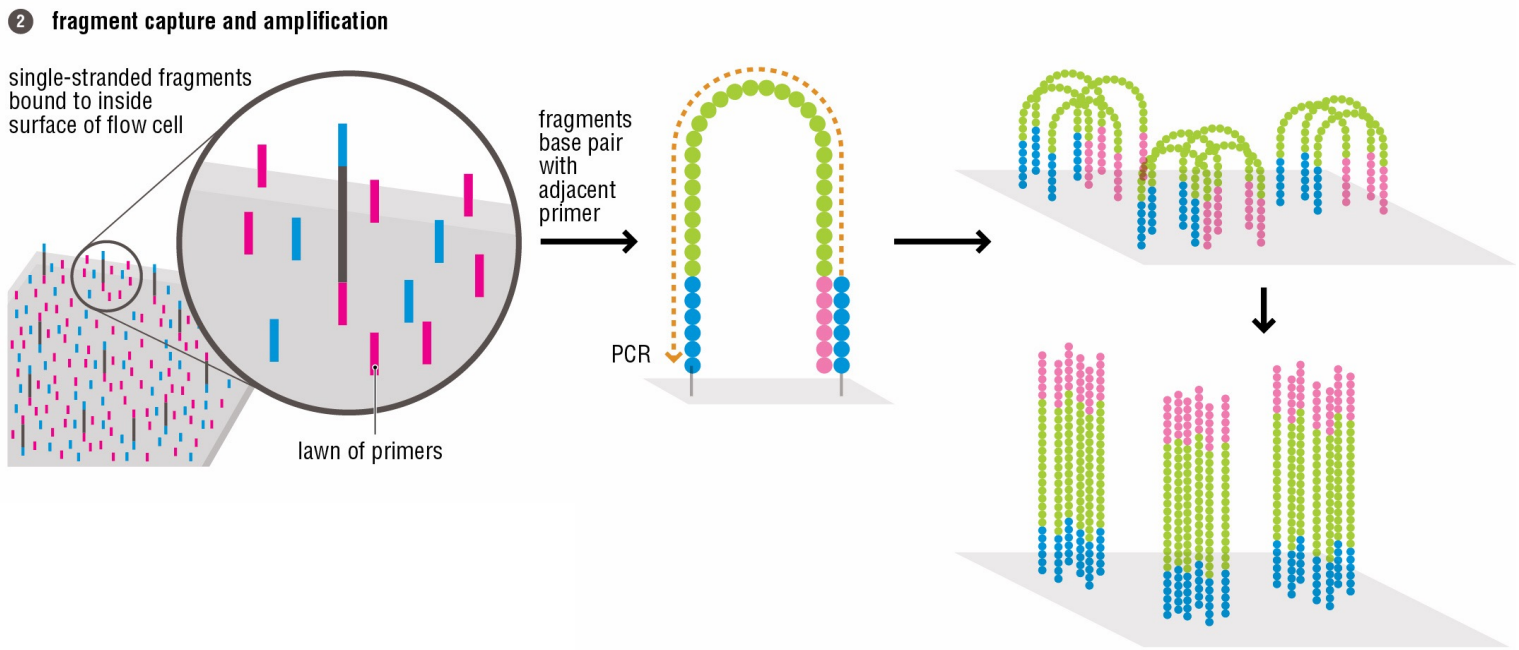
7. Data analysis





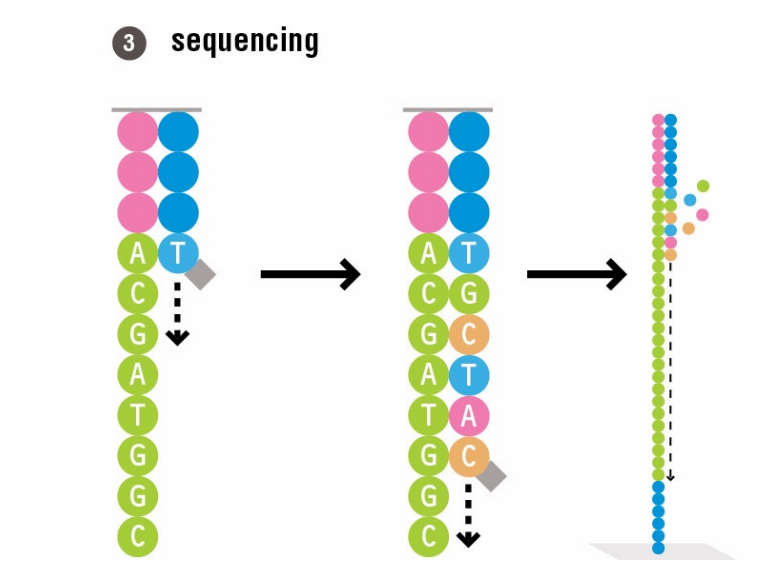
Illumina

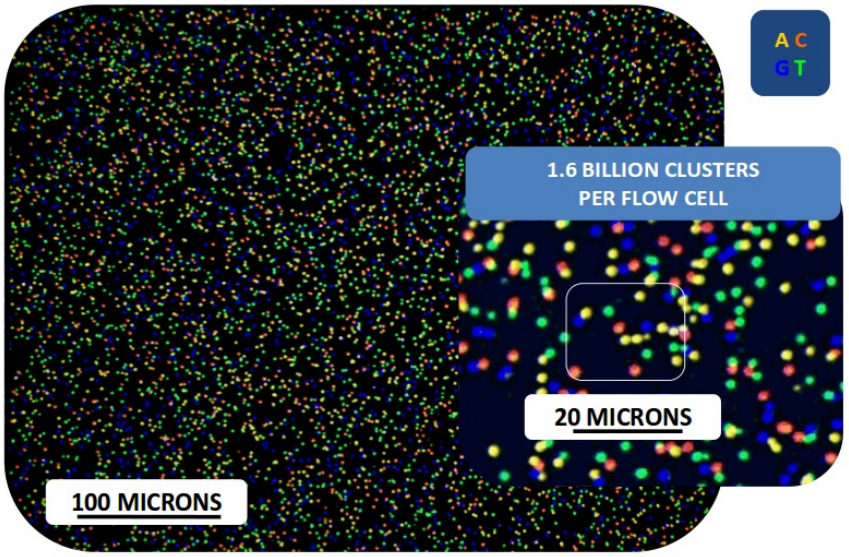
Sequencing by synthesis



Craig et al., Molecular Biology; Figure 19.94

Sequencing by synthesis

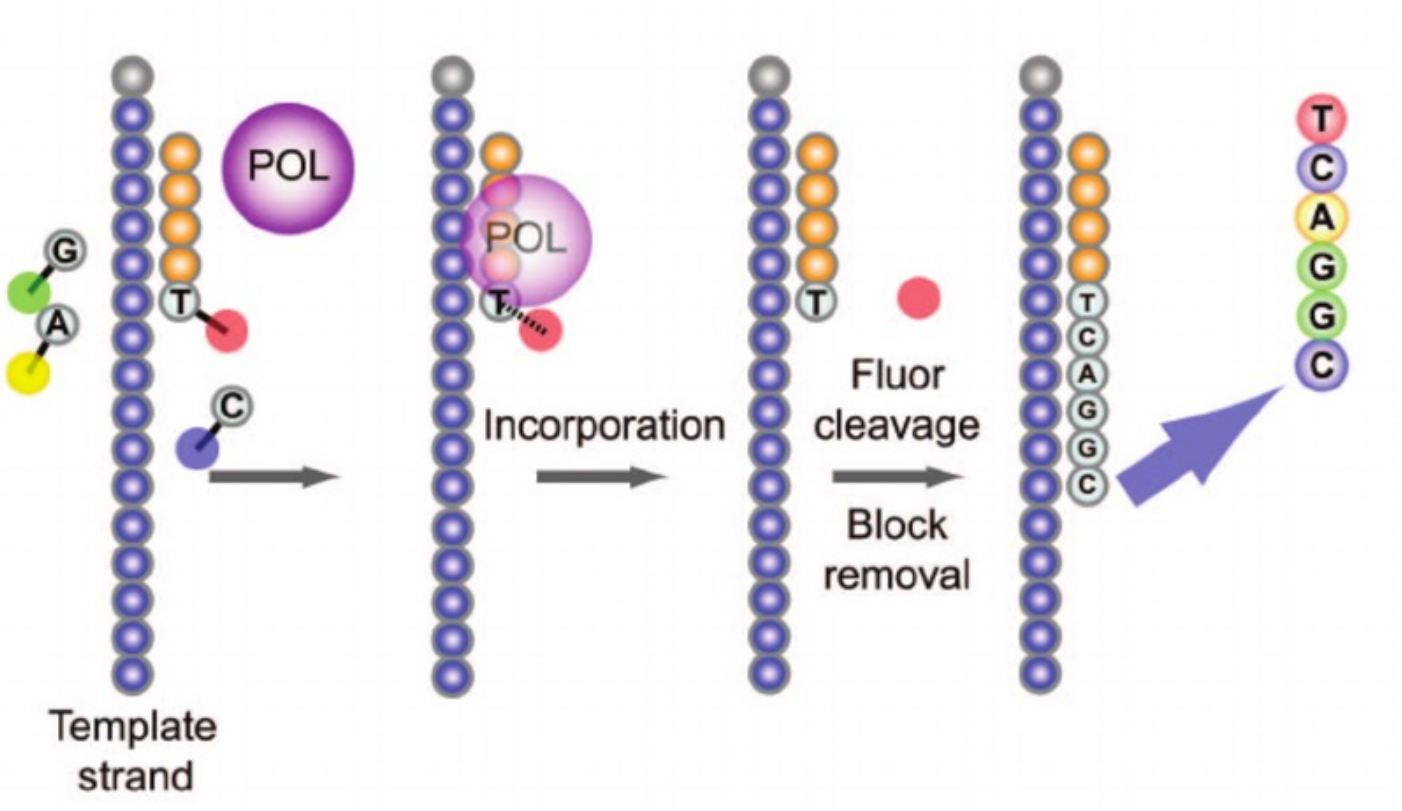




Craig et al., Molecular Biology; Figure 19.94

Sequencing by synthesis

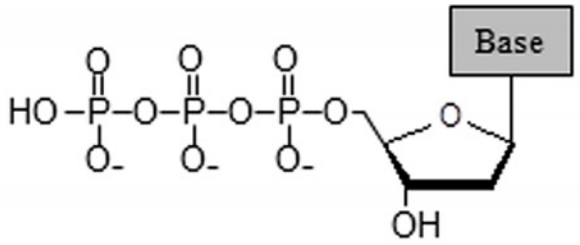
Reversible dye termination incorporates one fluorescently labeled nucleotide at a time



Sequencing by synthesis

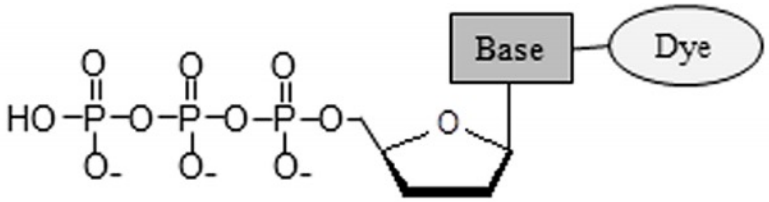
Reversible dye terminators

A



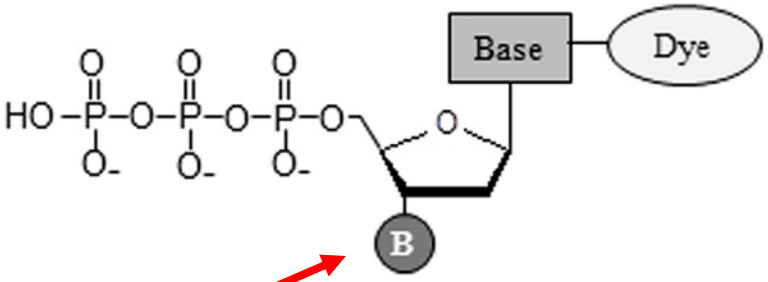
A regular deoxynucleotide (dNTP)

C



A dye-labeled dideoxy nucleotide (ddNTP) used in Sanger sequencing

D



removable dye

A reversible dye terminator used in Sequencing By Synthesis

removable blocking group

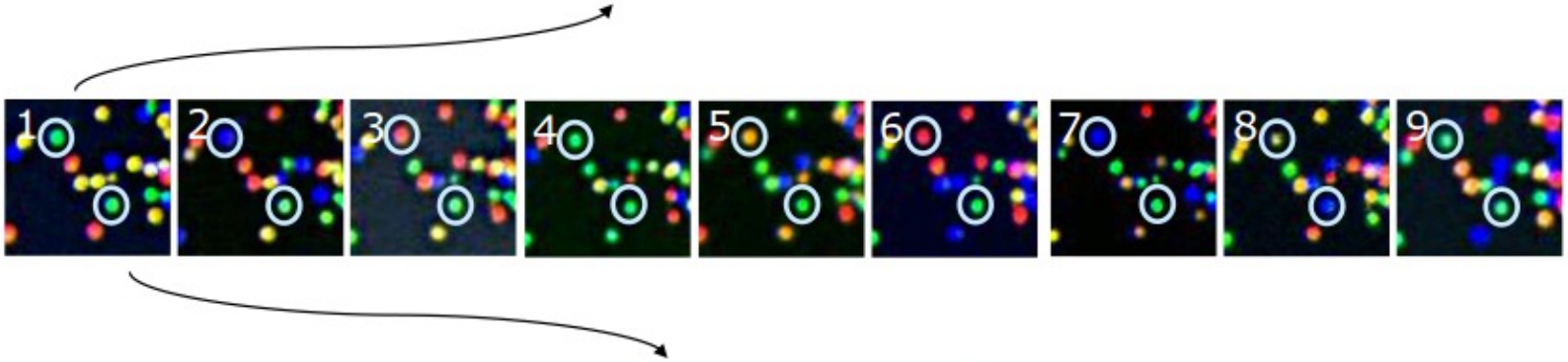
Chen, CY. 2014. Frontiers in Microbiol. Vol 5. Article 305.



Sequencing by synthesis

Quiz Yourself

ATGC





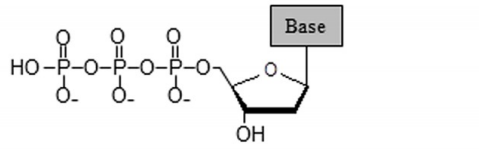
Sequencing by synthesis

Results will go here

Sequencing by synthesis

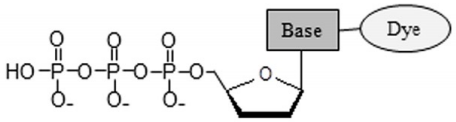
Quiz Yourself

A



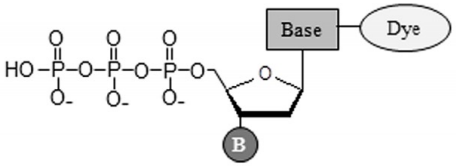
A regular deoxynucleotide (dNTP)

C



A dye-labeled dideoxy nucleotide (ddNTP) used in Sanger sequencing

D



A reversible dye terminator used in Sequencing By Synthesis

- What would happen if the sequencing tech accidentally substituted dNTPs for reversible dye terminators?
- What would happen if Sanger sequencing dye-labeled ddNTPs were used?



Sequencing by synthesis

Quiz Yourself

If Gene A is highly expressed and Gene B is lowly expressed, what do you expect?

A. The clusters from Gene A's fragments will be larger
B. The clusters from Gene A's fragments will be brighter
C. The reads from Gene A fragments will be longer
D. There will be more numerous clusters of Gene A's fragments



Decisions, Decisions



- Amount of input RNA?
- Method of mRNA enrichment?
- Strand specific (directional)?
- Type of library prep kit?
- Type of sequencing
  - Single-end or paired-end?
  - Read length, aka **cycles** (2x150)?
  - How many reads (> 20 million)?
  - What platform (novaSEQ or nextSEQ)?

ATGCGGTTATAGTATA
TGCCACCTAATTTCCC
CAGGTCCCCTTTGTTA
ATGCGGTTATAGTATA
TGCCACCTAATTTCCC

Resulting sequences

What do we get out?



Single-end sequencing

- 150 bp sequenced on one side of the insert
- Depending on the organism & project, 10,000,000 – 200,000,000 inserts sequenced
- 100 MB – 1 GB text file



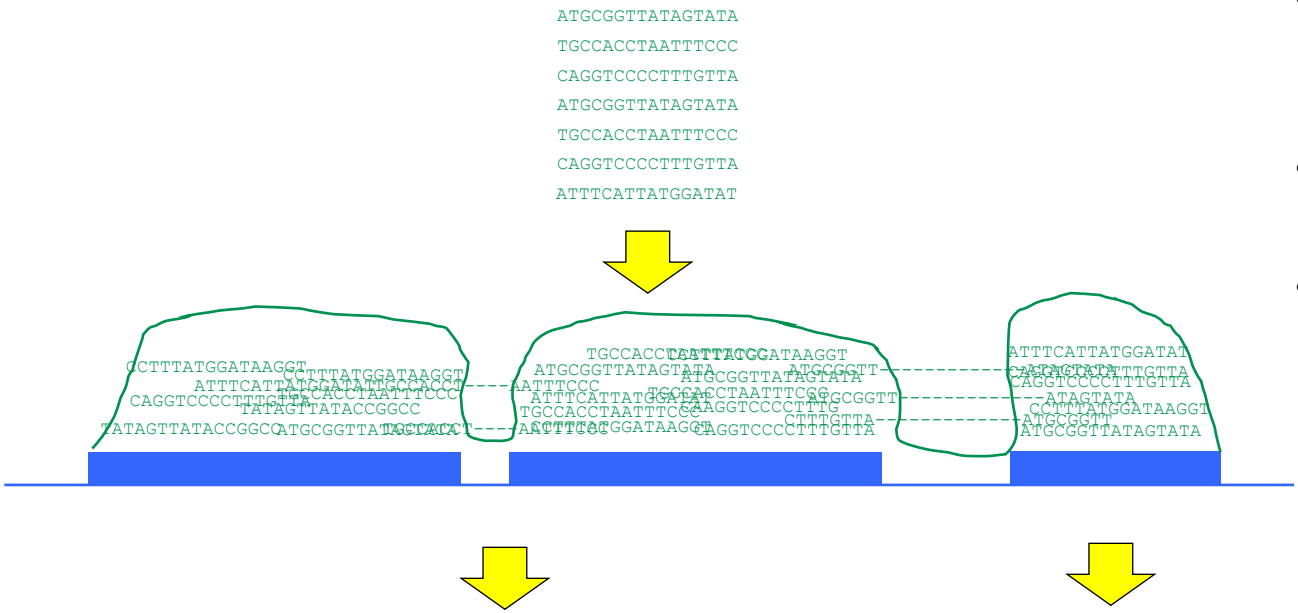
Paired-end sequencing

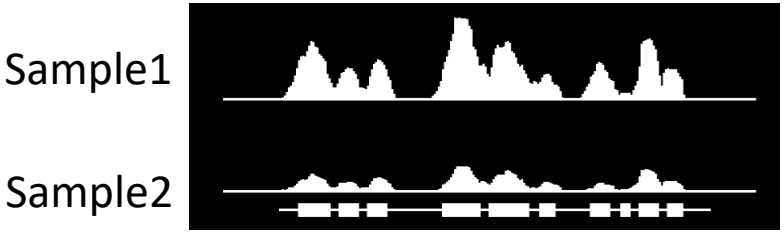
- 150 bp sequence on each side of the insert
- Depending on the organism & the project, 10,000,000 – 200,000,000 inserts sequenced
- Two 100 MB – 1 GB text files

ATGCGGTTATAGTATA
TGCCACCTAATTTCCC
CAGGTCCCCTTTGTTA
ATGCGGTTATAGTATA
TGCCACCTAATTTCCC

Data Analysis

Alignment





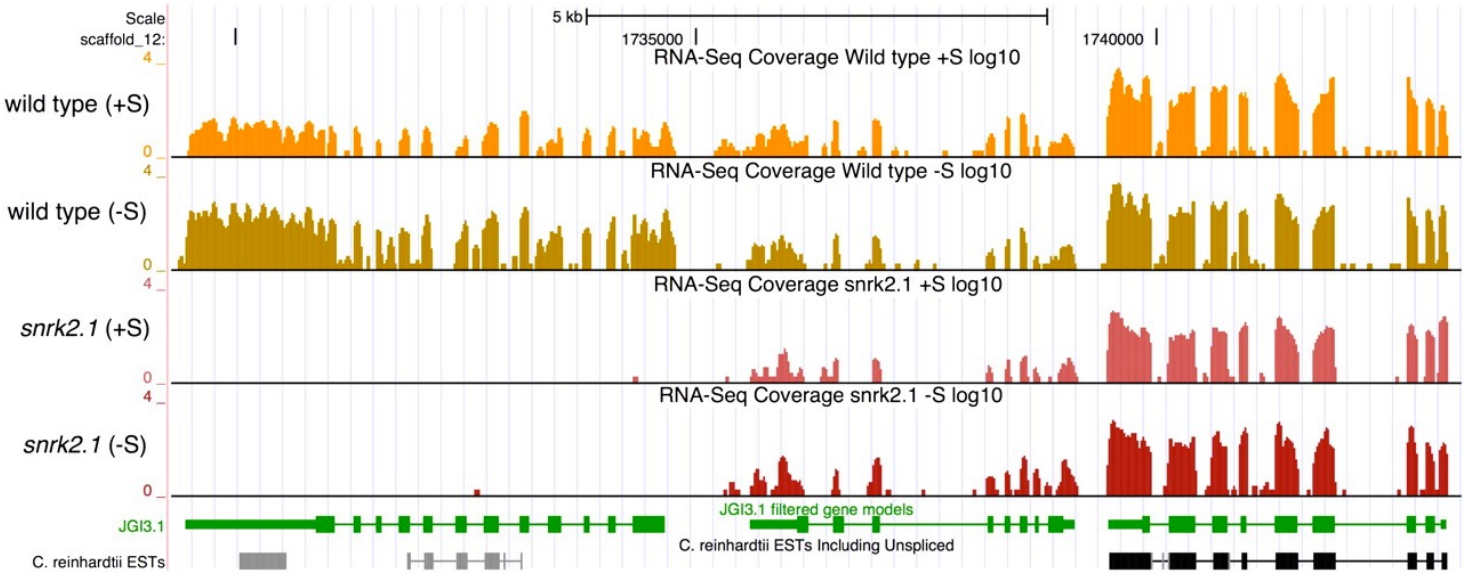
	Sample1	Sample2	…
Gene1	0	0	
Gene2	0	0	
Gene3	112	230	
Gene4	60	1003	
Gene5	0	0	
Gene6	3	0	
Gene7	234	360	
…			

- Sequenced fragments (called reads) will align to the exons
- There will be gaps at introns
- The number of reads mapped to a gene is proportional to its expression level (provided that some normalization has been performed)

ATGCGGTTATAGTATA
TGCCACCTAATTTCCC
CAGGTCCCCTTTGTTA
ATGCGGTTATAGTATA
TGCCACCTAATTTCCC

Genome browsers

See your data!

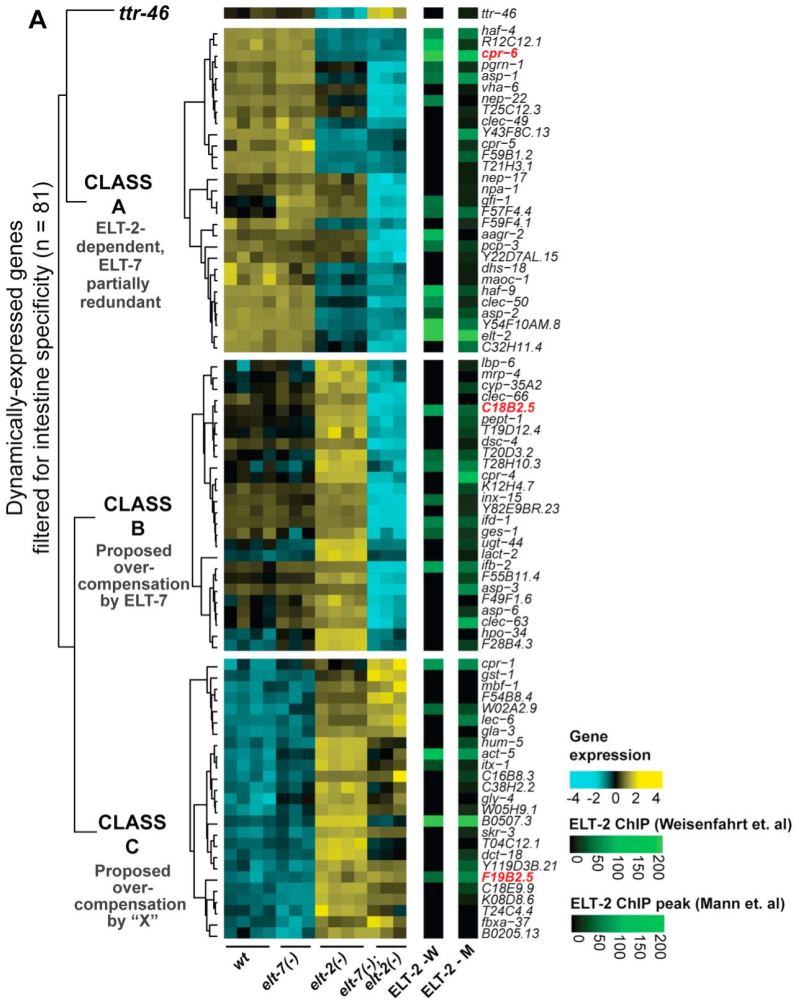


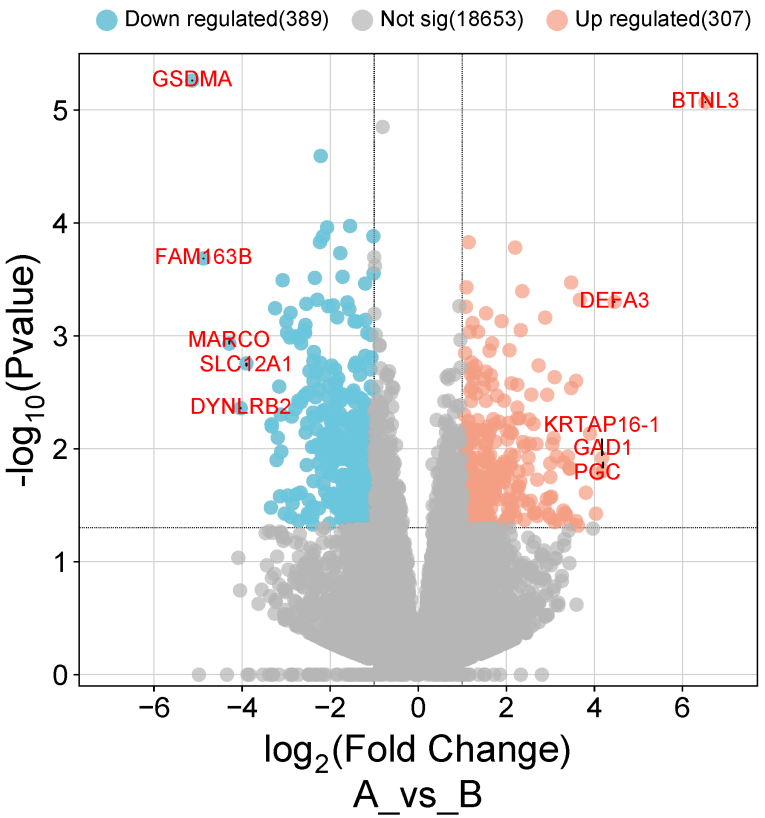
- Genome is below. Gene models are below (3 genes are shown).
- Samples are rows (two wild-types and two mutants are shown).
- Track heights represent the number of aligned reads for that region.
- Click around on a web-interface for the whole genome

ATGCGGTTATAGTATA
TGCCACCTAATTTCCC
CAGGTCCCCTTTGTTA
ATGCGGTTATAGTATA
TGCCACCTAATTTCCC

Beautiful Plots!

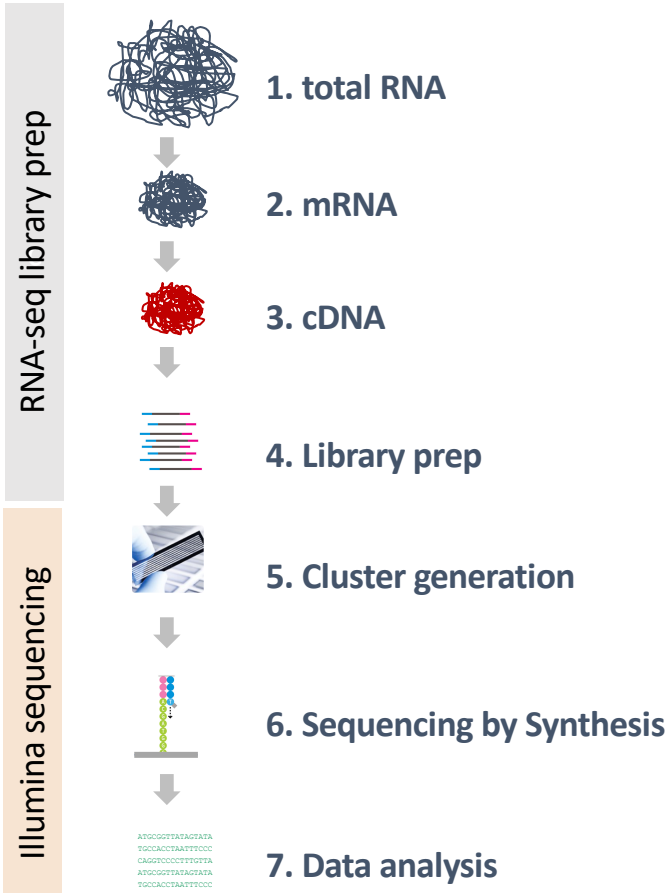
Publish and Share Your Results!





Illumina RNA-seq

Basic protocol recap



- Extract total RNA
- Enrich for mRNA
- Convert RNA to cDNA using Reverse Transcriptase
- Prepare libraries (fragmentation + adapter ligation)
- NEXTGEN sequencing is done at a facility. Samples are loaded onto a flow cell
- Sequencing by synthesis is performed to generate the sequence
- Data is sent back to researcher for alignment, quantification, and downstream analysis